

Multiple Choice (10 marks)

1. Find the Laplace transform $L\{g(t)\}$, where $g(t) = 0, t < 4$ and $g(t) = (t - 4)^2, t \geq 4$
 $(2 e^{-4s})/s^3$
 - (a) $2 / s^3$
 - (b) $(2 / s^2) * e^{-4s}$
 - (b) $4 / s^3$ $(2 e^{-4s})/s^4$
2. A box contains 7 red, 5 white, and 4 black balls. What is the probability of your drawing at random one red ball? One black ball?
 - (a) $P(R) = 2/3, P(B) = 1/3$
 - (b) $P(R) = 5/14, P(B) = 4/14$
 - (c) $P(R) = 1/2, P(B) = 1/4$
 - (d) $P(R) = 7/15, P(B) = 4/15$
 - (e) $P(R) = 7/16, P(B) = 1/4$
3. LC oscillators are used to produce a waveform with frequency ranging from
 - (a) 1 kHz to 100 kHz
 - (b) 500 MHz to 1 GHz
 - (c) 1 Hz to 1 kHz
 - (d) 1 KHz to 1 MHz
 - (e) 10 kHz to 100 MHz
4. An isolated spherical conductor of radius R carries a surface charge density σ . Find the potential energy in terms of R .
 - (a) $\{Q^2 / (16\pi\epsilon_0 R)\}$
 - (b) $\{2 Q^2 / (8\pi\epsilon_0 R)\}$
 - (c) $\{Q / (8\pi\epsilon_0 R)\}$
 - (d) $\{Q / (16\pi\epsilon_0 R^2)\}$
 - (e) $\{Q / (4\pi\epsilon_0 R^2)\}$
5. Find the attenuation per meter along a wave guide for an applied wave length $\lambda_0 = 2$ meters, if the cutoff wave length of the guide is $\lambda_{oc} = 20$ cm.
 - (a) 400 db/meter

- (b) 545 db/meter
- (c) 600 db/meter
- (d) 325 db/meter
- (e) 250 db/meter

True / False (10 marks)

Answer True or False

6. True or False: The maximum effective aperture of a short dipole is $0.075 \lambda^2$.
6. True or False: A reverse-biased diode can act as an amplifier by allowing some current to flow through the junction.
7. True or False: The inverse Laplace transform $L^{-1}[1/\{s(s^2 + 4)\}]$ is equal to $(1/2)\sin(2t)$.
8. True or False: A sinusoidal signal is a digital signal because it only has two values: positive and negative.
9. True or False: The electric field in the gap between two infinite parallel plates with equal surface charge density of the same polarity is zero.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given an air flow condition where the Mach number upstream of a shock is $M_x = 1.8$ and the static temperature downstream is $T_y = 800$, calculate the change in velocity across the shock. *D*
11. An induction motor operates with a rotor current of 10 amperes at a slip of 5 percent and a rotor effective resistance of 0.1 ohm per phase. Calculate the internal power developed per phase and discuss the implications of your findings on motor performance.

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